

Assignment 5 Interactive Data Visualization

STAT141B Spring 2024

Professor Duncan Temple Lang

Due: June 13, 10pm - ** NO EXTENSIONS **

Submit via Canvas

Your task is to create an interactive HTML page with interactive plots.

The page will have some text describing to the viewer

- what they are seeing,
- how to interact with the elements, and
- also a description of one or more insights from you exploring the data, e.g., guiding them towards some things to explore and how to find the insights.

We'll use the [data set](#) from the 600 rental units I collected from craigslist from Sacramento June 3rd. This is an RDS file that you can read into R via `readRDS()` after downloading the file to your computer.

Below the introductory text on the HTML page you create, include

- a map/plot showing each rental unit (apartment, condo or house) via its latitude and longitude values,
- a scatter-plot of square-footage and price,
- a 2 column table that shows the details of a given rental unit, i.e., column of field descriptions and a column of the corresponding field values.
 - The contents of this table will be displayed when the user clicks on a rental unit in the “map” plot.

Color code the points on the plots to indicate the value of a 3rd variable, e.g., price, number of bedrooms, number of bathrooms.

Customize the tooltips on the plots to show more information including a link to the post for the rental unit that the user can click to visit that page in a separate tab/window.

When one clicks on a point in the map,

- the corresponding point in the scatter-plot should become highlighted, and
- the table should show the details for the corresponding rental unit.

When one clicks on a different point, the previous point in the scatter-plot should become unhighlighted.

You can use R to create the plots and/or the HTML.

You can add JavaScript when creating the document from R via `saveWidget` or in a separate programmatic step.

You can also create the content directly in HTML and JavaScript.

You can generate HTML in R at page-creation-time and also in JavaScript at run-time.

As you write JavaScript functions, ask lots of questions. We want to do the basics and not focus on the language.

Keep It Simple (*KISS principle*)

This is a simple use of interactivity to learn the fundamentals. It is not intended to be very sophisticated or rich in user interface/experience details or functionality.

Focus on the mechanisms to get the interactivity and focus less on making the plots and page pretty. (Do make certain to put labels on each axis, legend, etc.)

Don't forget to make use of the Developer Tools in the Web browser to observe the interactive page and its elements and the output via the Console.

Strategy

I suggest working first on the longitude/latitude “map” and making each point clickable so that it displays the details of that post as an HTML table.

Then add the second plot. Then connect it to the events for the first plot.

Bonus

Make both plots interactive so that the user can click on a point in either plot and the points and the table are updated.

Possibly useful packages

- plotly
- crosstalk
- ggiraph
- patchwork
- htmlwidgets
- htmltools
- leaflet

Some possibly useful functions

- `ggplotly()`, `plot_ly()`
- `highlight()`, `highlight_key()`
- `HTML()`, `tags$...`, e.g., `tags$script()`, `tags$a()`, `tags$table()`
- `prependContent()`, `appendContent()`
- `onRender()`
- `style()`
- `subplot()`
- `layout()`
- `saveWidget()`

Links

- [JavaScript tutorial](#)
- [Plotly book](#)
 - [Chapter 16 - linking plots](#)
 - [Chapter 20 - Event Handlers](#)
 - [Chapter 21 - Custom Data](#)
 - [Chapter 10 - Saving widgets/pages](#)
 - [Chapter 25 - Tooltips](#)

Useful JavaScript Information/Functions/...

- `document` - the DOM object for the page.
- `getElementById()` - get the element with the specified id attribute.
- `innerHTML` attribute - specify the HTML content for a given DOM object which will be displayed when the attribute is set.